

Title

Model Evaluation & Performance Improvement Report

1. Introduction

This project focuses on evaluating multiple machine learning models using classification metrics and comparing their performance to identify the best model.

2. Models Used

- Logistic Regression
 - Decision Tree
 - Random Forest
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3. Evaluation Metrics

The following metrics were used:

- Accuracy → Overall correctness of model
 - Precision → Correct positive predictions
 - Recall → Ability to detect positives
 - F1 Score → Balance between precision and recall
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4. Model Comparison

A comparison was made between all models based on evaluation metrics.

Random Forest showed better performance compared to other models due to its ability to handle non-linear relationships.

5. Best Model Selection

The best model was selected based on **F1 Score**, as it provides a balance between precision and recall.

6. Conclusion

Model evaluation helped identify the most suitable model for the dataset. Random Forest performed best due to higher accuracy and robustness.